

NAME: _____

PERIOD: _____

Two Samples, One Difference

AP Statistics · Unit 4 · Topic 4.6

[STAR] THE PROBLEM

Suppose parcel pickup waits have the shown population parameters, measured in minutes. There are 600 North visits and 1,000 South visits. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE — one sentence before you work the parts

Would you check each place’s sample size separately or add them together? Give one reason from the study.

[BOOK] JUSTIFY EACH SAMPLE MEAN BEFORE THEIR DIFFERENCE (keep on the page)

For independent random samples, $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$ when each sample has $n \leq 0.10N$. Justify each component’s shape: a normal population supports a normal sample mean even for small n , while the Central Limit Theorem supports a sample mean from a nonnormal population when its own $n \geq 30$. The two groups need not use the same pathway. Interpret probabilities as outcomes from repeated pairs of samples. For a normal probability, standardize with $z = (\text{value} - \text{mean})/\text{SD}$ and use the appropriate normal tail. Printed normal table: $P(Z \leq 1.9220) = 0.972696$. A right-tail area is 1 minus the left area.

Work (a)–(c).

DOK 1 (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

DOK 2 (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

DOK 3 (c) In 4–5 sentences, decide whether 0.0273 is supported and explain what each student overlooks. Justify each sample mean using its own shape and size. Then consider $n_N = 48$ and $n_S = 8$: explain why their total alone cannot justify the same normal-model claim.

Frame: North uses _____; South uses _____.

Frame: With samples of 48 and 8, the total is _____, but
_____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.